

AI-Based Freelance-Startup System Progress Presentation

Week 06

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CATALOGUE

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01 Project Positioning

Current Phase

Phase 1: Intelligent Foundation

Focus: System Design and Core Architecture

Goal of Week 06: Link all components together to get a real app

Current Phase

Phase Tasks:

1. Database Modeling
2. AI Recommendation Tool (Candidate Match)
3. Profile Analyzer Tool
4. AI Predictive Tool
5. Selection and Mission Update
6. Mission Deployment
7. Application Tracking System
8. Profile Engineering and Authentication



Phase Tasks:

1. Database Modeling
2. AI Recommendation Tool (Candidate Match)
3. Profile Analyzer Tool
4. Selection and Mission Update
5. Mission Deployment
6. Application Tracking System
7. Dual-Sided AI Match Alerts
8. Profile Engineering
9. Performance Score (Freelancer Side)
10. Contract Generation
11. Sign-in / Sign-up (JWT Authentication)

Weekly Objective

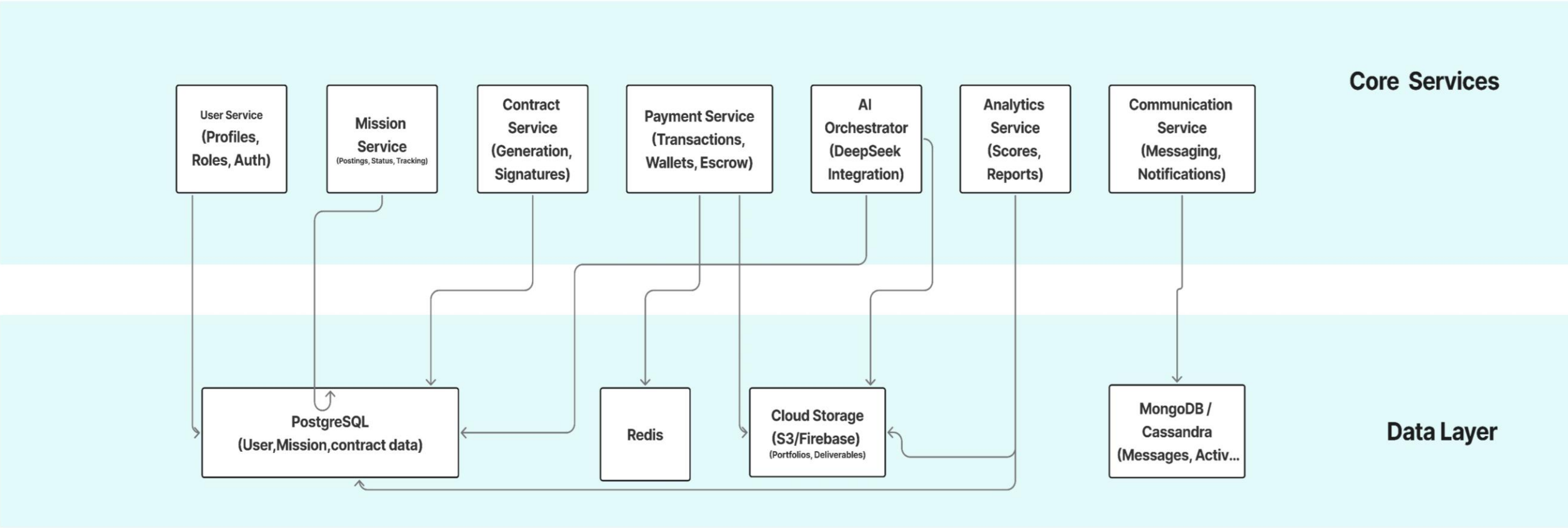
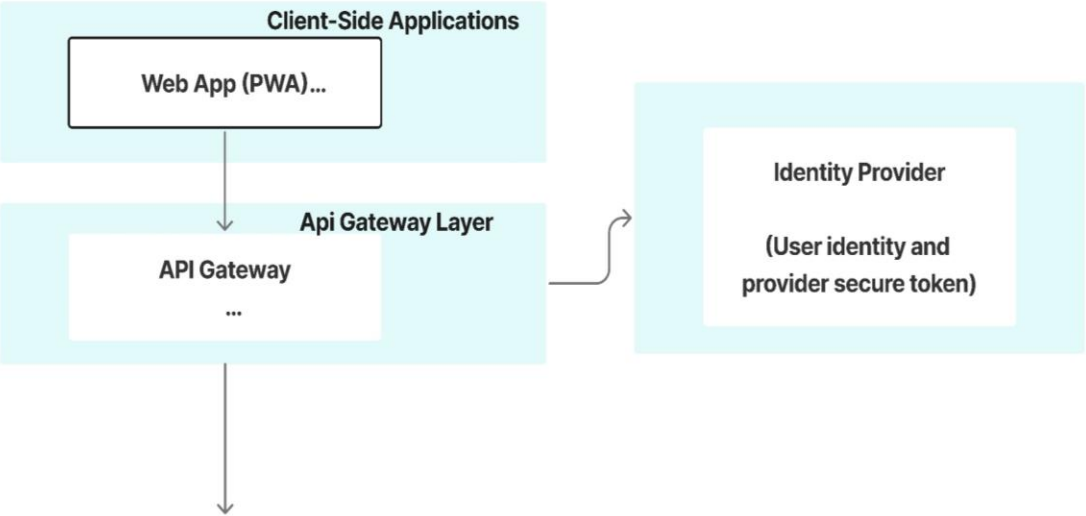
- 1 Integrate all Phase 1 components into a fully functional system (main objective).
- 2 Launch Phase 2 and Phase 3 concurrently to accelerate overall development.



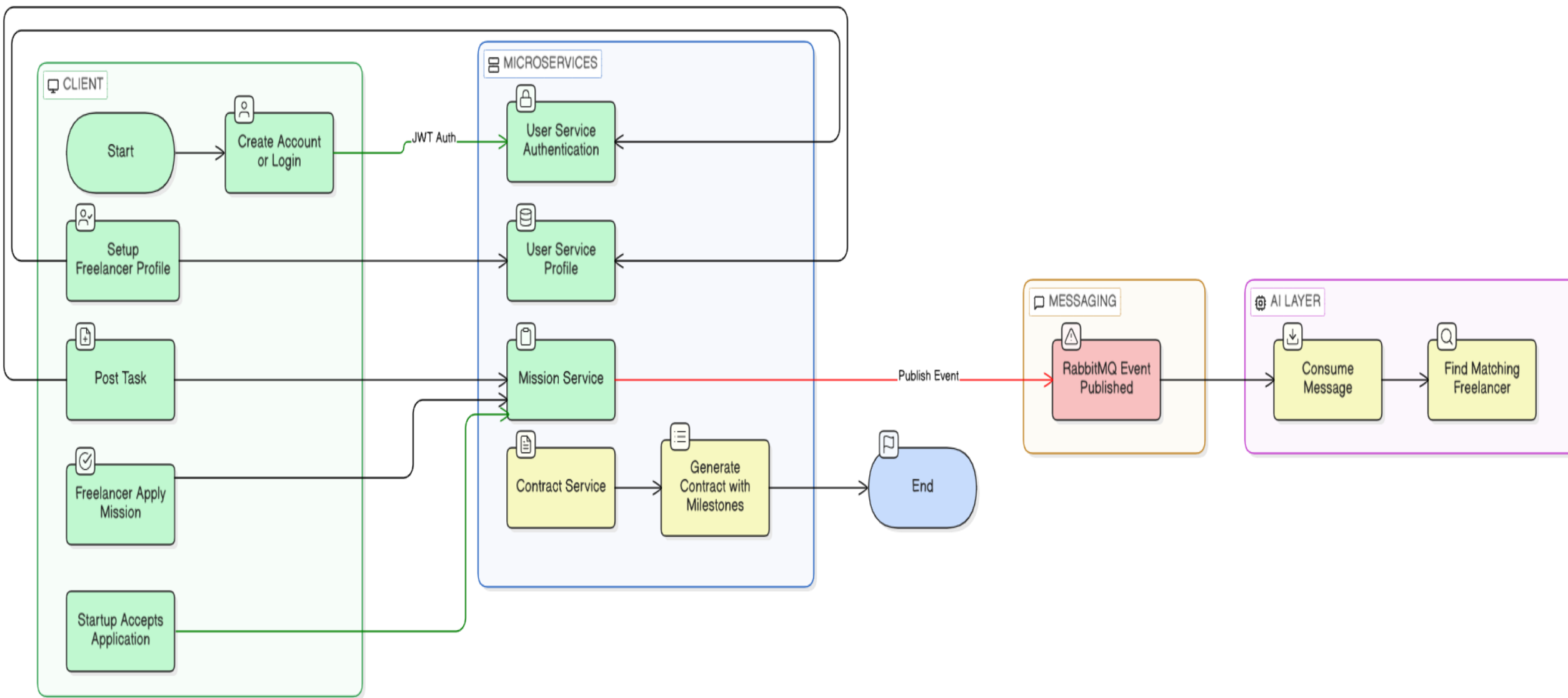
02

Weekly Progress

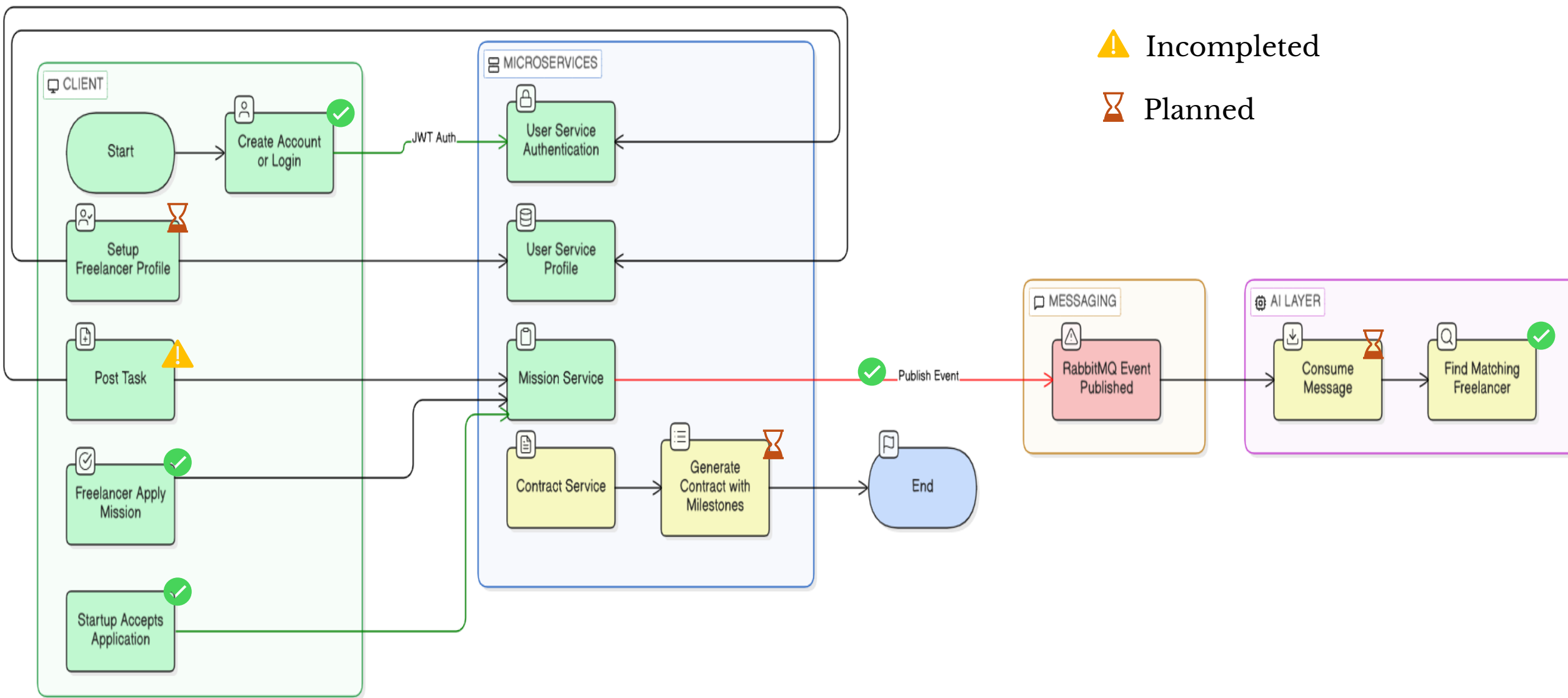
Project Architecture



Project Architecture (Phase 01)



Project Architecture (Phase 01)



Project Architecture (Phase 01)

Phase Tasks:

Database Modeling 

AI Recommendation Tool (Candidate Match) 

Profile Analyzer Tool 

Selection and Mission Update 

Mission Deployment 

Real -Time notification system 

Profile Engineering 

Performance Score (Freelancer Side) 

Contract Generation 

 Done

 Incompleted

 Planned

2 AI and Intelligent Layer

2.1 Integrated Tools

1. Freelancer Tools

search freelancer
get freelancer profile

2. Company and Mission Tools

get company profile
search mission
get mission details

3. AI Matching and Analysis

calculate compatibility score
find best matching mission
get matching missions

4. Advanced Deep Analyzers

analyser for freelancer profile

2. AI and Intelligent Layer

2.2 Architecture and Methodology

RAG system (Agentic):

Standard AI, decides which platform "tools" to run (querying PostgreSQL, calculating scores, fetching analytics) to provide factual, data-driven responses.

Event-Driven Matching (EDA):

Integrated with RabbitMQ, When a startup posts a job, the AI Orchestrator automatically wakes up, scans the entire freelancer pool, and pushes recommendations back to the ecosystem without any human intervention.

Weighted Compatibility Scoring:

It uses a multi-dimensional scoring algorithm that evaluates candidates based on:

- * Skill Overlap and Proficiency: (35%)
- * Rate Alignment: (25%)
- * Experience Level: (25%)
- * Success Rate/History: (15%)

2. AI and Intelligent Layer

2.2 Architecture and Methodology

Automated Matching Workflow :

1. **Event Trigger:** A Company publishes a mission » ``mission-service`` emits a ``mission_published`` message to RabbitMQ.
2. **Ingestion:** ``worker.py`` receives the message and fetches the full mission details (skills, budget, requirements).
3. **Discovery:** The service queries the database to find freelancers whose skills overlap with the mission requirements.
4. **Scoring:** Each potential candidate is passed through the **AI Scorer** (``database.py``) to calculate a match percentage (0-100%).
5. **Recommendation:** Candidates with a score above 20 are sorted and sent back to RabbitMQ.
6. **Notification:** The platforms' WebSocket service picks up these matches and notifies the Freelancer: **"We found a mission that fits your profile!"**

Key Technologies :

- * **FastAPI:** High-performance Python web framework.
- * **RabbitMQ (aio-pika):** Asynchronous message brokering for the event-driven worker.
- * **PostgreSQL:** Primary data source for the matching engine.
- * **Llama 3 :** Large Language Models used as the reasoning core

2. AI and Intelligent Layer

2.3 Matching Test Screenshots

Bridge-Project docker exec bridge-postgres psql -U postgres -d FreelaceStartupsPlatform -c "SELECT id, title FROM project_service.missions LIMIT 1;"

id	title
aaaa0000-0000-0000-0000-000000000001	Full-Stack Web Application Development

(1 row)

Bridge-Project docker exec bridge-rabbitmq rabbitmqadmin publish \

exchange=mission_exchange \

routing_key=mission_published_key \

payload='{"missionId":"aaaa0000-0000-0000-0000-000000000001"}' \

properties='{"content_type":"application/json","delivery_mode":2}'

Message published

Bridge-Project docker exec bridge-postgres psql -U postgres -d FreelaceStartupsPlatform -c "SELECT id, title FROM project_service.missions;"

id	title
aaaa0000-0000-0000-0000-000000000001	Full-Stack Web Application Development
aaaa0000-0000-0000-0000-000000000002	Machine Learning Pipeline for Patient Data Analysis
aaaa0000-0000-0000-0000-000000000003	Cross-Platform Mobile App Development
aaaa0000-0000-0000-0000-000000000004	Data Analytics Dashboard
44713944-cade-460a-9909-e5f8b5d5bd2b	tt
a76b5e20-023a-44a5-befa-96bbfb3e90d4	FullStack devs for Saas project
3be6d64e-2c92-4967-a79d-b900f65796db	vvvv
cb5d5fb3-5cc9-406c-8c1e-910b4492580f	xxx
b2caa50b-df7d-440d-9954-33e23875bcf7	cccc

(9 rows)

Bridge-Project docker exec bridge-rabbitmq rabbitmqadmin publish \

exchange=mission_exchange \

routing_key=mission_published_key \

payload='{"missionId":"a76b5e20-023a-44a5-befa-96bbfb3e90d4"}' \

properties='{"content_type":"application/json","delivery_mode":2}'

Message published

2. AI and Intelligent Layer

2.3 Matching Test Screenshots

```
bridge-ai-orchestrator | INFO: 127.0.0.1:40060 - "GET /health HTTP/1.1" 200 OK
bridge-ai-orchestrator | INFO: 127.0.0.1:50854 - "GET /health HTTP/1.1" 200 OK
bridge-ai-orchestrator | INFO: 127.0.0.1:59776 - "GET /health HTTP/1.1" 200 OK
bridge-ai-orchestrator | INFO: 127.0.0.1:43562 - "GET /health HTTP/1.1" 200 OK
bridge-ai-orchestrator | 2026-04-13 22:40:21,236 - app.utils.worker - INFO - [Worker] 📄 New mission published: aaaa0000-0000-0000-0000-000000000001 - running AI match ...
bridge-ai-orchestrator | 2026-04-13 22:40:21,257 - app.utils.worker - INFO - [Worker] Mission 'Full-Stack Web Application Development' loaded.
bridge-ai-orchestrator | 2026-04-13 22:40:21,299 - app.utils.worker - INFO - [Worker] Scoring 5 candidate(s) ...
bridge-ai-orchestrator | 2026-04-13 22:40:21,311 - app.utils.worker - INFO - [Worker] ✓ 5 match(es) found for mission aaaa0000-0000-0000-0000-000000000001. Top score: 92.5
bridge-ai-orchestrator | 2026-04-13 22:40:21,518 - app.utils.worker - INFO - [Worker] 📄 Published 5 recommendation(s) for mission aaaa0000-0000-0000-0000-000000000001 → 'ai_recommendations_queue'
bridge-ai-orchestrator | INFO: 127.0.0.1:50402 - "GET /health HTTP/1.1" 200 OK
bridge-ai-orchestrator | INFO: 127.0.0.1:38064 - "GET /health HTTP/1.1" 200 OK
bridge-ai-orchestrator | INFO: 127.0.0.1:45264 - "GET /health HTTP/1.1" 200 OK
bridge-ai-orchestrator | INFO: 127.0.0.1:43324 - "GET /health HTTP/1.1" 200 OK
bridge-ai-orchestrator | INFO: 127.0.0.1:43816 - "GET /health HTTP/1.1" 200 OK
bridge-ai-orchestrator | INFO: 127.0.0.1:37868 - "GET /health HTTP/1.1" 200 OK
bridge-ai-orchestrator | INFO: 127.0.0.1:52290 - "GET /health HTTP/1.1" 200 OK
bridge-ai-orchestrator | INFO: 127.0.0.1:52972 - "GET /health HTTP/1.1" 200 OK
bridge-ai-orchestrator | INFO: 127.0.0.1:41628 - "GET /health HTTP/1.1" 200 OK
bridge-ai-orchestrator | INFO: 127.0.0.1:41858 - "GET /health HTTP/1.1" 200 OK
bridge-ai-orchestrator | INFO: 127.0.0.1:35168 - "GET /health HTTP/1.1" 200 OK
bridge-ai-orchestrator | INFO: 127.0.0.1:57714 - "GET /health HTTP/1.1" 200 OK
bridge-ai-orchestrator | INFO: 127.0.0.1:48886 - "GET /health HTTP/1.1" 200 OK
bridge-ai-orchestrator | INFO: 127.0.0.1:42502 - "GET /health HTTP/1.1" 200 OK
bridge-ai-orchestrator | 2026-04-13 22:47:34,066 - app.utils.worker - INFO - [Worker] 📄 New mission published: a76b5e20-023a-44a5-befa-96bbfb3e90d4 - running AI match ...
bridge-ai-orchestrator | 2026-04-13 22:47:34,087 - app.utils.worker - INFO - [Worker] Mission 'FullStack devs for Saas project' loaded.
bridge-ai-orchestrator | 2026-04-13 22:47:34,091 - app.utils.worker - INFO - [Worker] Scoring 8 candidate(s) ...
bridge-ai-orchestrator | 2026-04-13 22:47:34,110 - app.utils.worker - INFO - [Worker] ✓ 8 match(es) found for mission a76b5e20-023a-44a5-befa-96bbfb3e90d4. Top score: 94.7
bridge-ai-orchestrator | 2026-04-13 22:47:34,184 - app.utils.worker - INFO - [Worker] 📄 Published 8 recommendation(s) for mission a76b5e20-023a-44a5-befa-96bbfb3e90d4 → 'ai_recommendations_queue'
bridge-ai-orchestrator | INFO: 127.0.0.1:38254 - "GET /health HTTP/1.1" 200 OK
bridge-ai-orchestrator | INFO: 127.0.0.1:59652 - "GET /health HTTP/1.1" 200 OK
bridge-ai-orchestrator | INFO: 127.0.0.1:50558 - "GET /health HTTP/1.1" 200 OK
bridge-ai-orchestrator | INFO: 127.0.0.1:39732 - "GET /health HTTP/1.1" 200 OK
```



03

Planning and Next steps

Plan for the Next Week

- 1 Integrate all Phase 1 components into a fully functional system (main objective).

Thank you